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Reaction Simulation (RS) Extension for SIMION®

Documentation / Manual

(RS version 0.4.4)

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1 Introduction

Welcome to the documentation of the Reaction Simulation (RS) extension¹ to the charged particle solver SIMION^{®2}. RS utilizes the Lua³ user programming interface of SIMION to simulate the chemical reaction dynamics of charged particles under atmospheric pressure (AP) conditions, which is a common problem in the design and investigation of modern ion analytical devices like ionization sources for mass spectrometry or ion mobility spectrometers.

1.1 Ion dynamics under atmospheric pressure conditions

Under ideal vacuum conditions the trajectories of charged particles depend solely on the start configuration of the ions (initial motion vector) and on the present electric and magnetic fields. One can assume such ideal conditions if the mean free path of a charged particle at a given background gas pressure is much longer than its path in an actual technical device. If an ideal vacuum can be assumed the charged particles can not chemically interact with other particles; Therefore only internal chemical transformations like rearrangements or fragmentation reactions can occur, depending on the internal energy the charged particles carry from preceding ionization process for example. If no vacuum is maintained, then charged particles, like molecular ions, experience collisions with bulk gas particles. Of course the collision frequency depends on the local background gas pressure. At atmospheric pressure (AP) conditions (1 bar background gas pressure), a typical collision frequency for small molecular ions is in the range of 10^9 s^{-1} , which corresponds to a mean free path of around 100 nm.

¹Additional information and the latest RS version can be found at <http://rs.ipams.uni-wuppertal.de>

²SIMION is a registered trademark of Scientific Instrument Services, Inc. <http://simion.com/>

³Lua is a multiparadigm scripting language <http://www.lua.org>

2 Method

Therefore at elevated gas pressures the motion of ions is obviously also dependent on the interaction with the bulk gas. At AP the viscous drag force and molecular diffusion acting on small molecular ions are readily comparable to the applicable electrical or magnetic forces. Thus at high pressures ions "drift" through a viscous environment instead of flying through an electromagnetic field. Some basics of the motion of ions under AP can be found in [1].

In addition to the effects on the trajectories, the chemical dynamics become drastically more complicated because the charged particles can now be chemically transformed by all possible reactions with any background gas component or contaminant.

1.2 Basic idea of RS

The basic idea of RS is to extend the existing methods for the simulation of charged particle trajectories under elevated pressure with a numerical model of the chemical reaction dynamics of the charged particles. The particle trajectory and reaction simulation are modularized. Basically the RS code could be used with any particle collision model (even unimolecular transformation reactions under ideal vacuum could be modeled with RS). However in the actual implementation, RS is coupled to the statistical diffusion simulation (SDS)[2] model, which describes ion trajectories under roughly atmospheric pressure conditions.

2 Method

2.1 Monte-Carlo simulation

In a monte-carlo type simulation a given problem is statistically described by many elementary random experiments. If the results of the elementary processes are analyzed appropriately, the modeled, often very complicated, larger problem can be described with a high statistical confidence. The benefit of monte-carlo type simulations is that the elementary processes are

typically relatively simple in comparison to general mathematical models of the whole process, which are often complex systems of partial differential equations. This allows often the quite simple simulation of complicated phenomena, which would be hard with other numerical methods like finite element (FEM) simulations.

In the case of RS the dynamics of a very large number of charged particles, like a whole package of 10^7 molecular ions generated by a single laser shot for example, is simulated by a much smaller group of simulated particles. Typical numbers of simulated particles would be in the range of 10^5 . The dynamics of the individual simulated particle is also simulated with a monte-carlo type approach: The extremely high number of gas collisions occurring at AP along an individual charged particle trajectory is approximately described by the SDS model, which performs only one representative random jump per time step. Similarly, the chemical reaction dynamics are described by discrete reaction events, also in one reaction phase per simulation time step, by the Reaction Simulation extension (RS).

If the described simulated group of sample particles follow the same statistics as the real physical ion package and the simulated sample has a sufficient size, the dynamics of the real charged particles can be correctly simulated by the described process.

2.2 Statistical diffusion simulation (SDS)

As described earlier the simulation process requires a model for the interaction of the charged particles with the background gas. The most simple straightforward method for such a model would be hard sphere scattering. In this case, ions and bulk gas particles are approximated as hard spheres and every single collision is individually simulated. Despite the simplicity of this approach (and its benefit in medium pressure regimes) at AP it is generally not applicable be-

cause the number of collision events would be too high to be handled by actual computer hardware.

The statistical diffusion simulation algorithm [2], developed by Appelhans and Dahl, treats the effect of the many collision events occurring per time step of the ion trajectory integration process of SIMION by substituting them with a single representative random jump. This random jump follows the same statistics as the net effect of the individual collisions. With this approach, the SDS algorithm is able to reduce the required numerical effort drastically, if the basic assumptions of SDS hold place. The main assumption is that the ion motion vector becomes completely decoupled between the simulation time steps. At atmospheric pressure, the number of collisions between the time steps is very high so that this assumption is certainly valid.

The basics of SDS can be found in [2], and examples of the application of SDS are [3],[4] or [5].

2.3 RS simulation process

The basic simulation process of RS is relatively simple. For every time-step of the SIMION ion trajectory integration the following tasks are performed for every simulated ion:

- Calculate the individual reaction probability p_R for the reaction R of the actual particle:

$$p_R = \prod_i (c_i) k_R dt \quad (1)$$

with the local concentration of the i -th neutral reaction partner c_i , the rate constant of the reaction k_R and the length of the timestep dt .

- Generate a uniformly distributed random number r in the interval $[0, 1]$. If r fulfills $r < p_r$ the reaction event is considered as actually happening. The educts of the reaction event are removed from the set of simulated particles and the products are added to it.

- If the reaction event did not happen, the process is repeated with the next reaction of the set of possible reactions of the actual particle (the set of reactions which have the actual particle type on the educt side).

Therefore to simulate the reaction dynamics of a given set of ions, one has to know the chemical kinetic data of the elemental reactions of those ions in the form of reaction equations with the according reaction rate constants.

2.4 Current restrictions

The current implementation of RS has some restrictions, which result from the relatively early current development stage and are not fundamentally tied to the simulation principle:

Isotropic and static reaction partner distribution

Currently RS assumes static and isotropic concentration distributions of the neutral reaction partners. Like the velocity, temperature and pressure distributions of SDS, such spatial resolved concentration fields could be provided in simple text files or (in SIMION 8.1) as PA files. The import of such files would be relatively straightforward with the existing Lua methods provided by SIMION / SDS. Therefore spatial resolved concentration fields are readily possible if needed.

No temperature dependence

The reaction conditions are assumed to be strictly isothermic. The simulated reaction temperature can be controlled by the rate constants. Therefore the user has to enter the correct rate constants for the intended simulation temperature.

Currently effects from electrical heating (i.e. effects from the increased kinetic energy due to electrical acceleration of the ions) are also neglected.

No pressure dependence

Similar to the assumption of isothermal conditions, RS currently assumes isobaric conditions. The background gas pressure is defined implicitly with the given reaction system, either with the number concentration of a neutral reaction partner (mostly the background buffer gas) or by the values of pressure dependent reaction rates. Like in the case of the concentration distribution of the reaction partners, a spatial resolved pressure field could be used to scale the number concentrations of neutral reaction partners.

Notice: If the pressure is changed in a RS simulation, the ratio between the number of simulation particles and the physical charged particles may change too. Depending on the details of the modeled system, the number of charged particles per volume unit may drop if the background pressure drops. If the number of simulation particles is not changed in such a situation, one simulation particle then represents less physical particles.

No ion-ion reactions

Currently RS does not support reactions between ions. To simulate such reactions the collisions between ions, which are simulated particles in the case of RS, has to be correctly modeled. This would complicate the simulation algorithm because now the distances between the simulated ions would have to be calculated in every time step to get the collision events between the simulated particles. This necessary distance search would significantly extend simulation runtimes. Because our group did not need the capability to simulate ion recombinations or similar reactions, we neglected it.

Due to current restrictions of the SIMION API, RS is also not capable to simulate reactions which result in two or more smaller fragment ions. An example of such a reaction would be a

coulomb explosion of a highly charged droplet from an electrospray. Because for this reaction type only one ionic species occurs on the educt side, it is not necessary to perform an exhaustive distance estimation between the simulated particles. If the SIMION API is updated, fragmentation reactions of ions in multiple charged sub-particles will be available by RS.

Notice: Reactions which have only one ionic species on the educt side are called *independent* (of other ionic species) in RS, if there is more than one ionic species on the educt side, the reaction is called *dependent*.

Warning: Dependent reactions are **silently ignored** by the simulation process, despite the fact that the config file parser will accept simulation configurations containing definitions of dependent reactions.

3 RS Usage

The following discussion gives an introduction into the usage of RS and the format of the simulation configuration file format.

3.1 SIMION integration / How-To

3.1.1 Basic setup:

RS is basically a SIMION user program written in Lua. It was tested with SIMION 8.0 and 8.1.

Notice: It may work in SIMION 8.0, but `collision_sds.lua` presently should be taken from the latest release of SIMION 8.1.x by clicking "Check for Updates" on the SIMION 8.1 main screen.

The RS folder contains an example SIMION project (`test.iob`), which shows the basic integration of RS with SIMION. As shown in the SIMION workbench user program of the test SIMION project (`test.lua`) the RS simulation

code can be included into a SIMION simulation by importing the main module of RS `RS_collision_sds.lua`:

```
-- Load SDS / RS user program:
local SDS = simion.import ↵
    ("RS_collision_sds.lua")
```

As the name of the main module suggests, this also imports a version of the statistical diffusion simulation (SDS) algorithm. RS is based on the SDS implementation in Lua, which is shipped with SIMION (since version 8.1.1.16). Details of the SDS algorithm and its configuration for a productive simulation can be found in the SDS documentation and in [2].

3.1.2 Running a simulation with RS

Preparations and ion start position initialization:

The configuration of RS for a simulation run is contained in a plain text configuration file named `RS.conf` (See chapter 3.3 for details of the config file format). To run a simulation the RS configuration file has to be updated with the intended simulation parameters.

Due to the lack of a free usable parameter of the ions in SIMION, RS utilizes the ion color index (an integer number) as the ion substance index. Therefore the assignment between ions in SIMION and their substances in RS is done via the ion color index. For example if it is intended to start the simulation run with a group of ions consisting of particles of the third substance in the RS substance list, those ions in SIMION must have the color index 3. The color index is updated during the simulation, therefore the ions will change their color in SIMION according to their changes by the simulated reactions.

Warning: The particle mass and the particle charge of the simulation particles are taken from the RS configuration file. The values assigned to the particles in SIMION are **silently overwritten** with the values from the RS configuration file.

Running the simulation If a simulation run is started ("Fly'm" in SIMION), the SDS algorithm and RS is initialized before ions are actually flown. SDS imports gas velocity, temperature and pressure fields from any existing parameter files (please refer to the SDS documentation for details of the file format) and initializes its internals. RS imports and parses its configuration file and also initializes its internal data structures (tables with substances and reactions in the simulation, a table of all simulated particles with links to their substances, etc.). After this initialization the common particle tracing integration loop of SIMION starts.

As stated in the introduction, RS runs the chemical reactions of every simulated particle in every time step of the SIMION particle tracing. This means that the SIMION integration time steps are also the time steps of the chemical simulation in RS. Thus they have to be short enough that the chemical simulation remains stable to ensure valid results. If the time steps (dt in equation 1) are chosen too long, the reaction probability of distinct reactions can become > 1 . In this case the whole simulation process can become unstable. A typical indicator for this situation is the occurrence of heavy artificial concentration oscillations in the RS simulation.

The maximum time step length possible for a valid simulation will depend on the details of the modeled chemistry, but for fast reaction systems the demands of RS in terms of maximum time step length are much higher than those of the SIMION particle migration simulation. For very fast chemical systems (the equilibrium of small water clusters at high water background levels for example) the required time step length is in the range of $0.005 \mu\text{s}$. Details of the controlling of the time stepping in a SIMION/SDS-RS simulation can be found in the SIMION documentation.

Fortunately the detection of "ill" reaction events with reaction probabilities $p > 1$ is of course easy. If an ill reaction event is detected RS writes a warning message to the log if the `RS_log_ill_events` adjustable variable is 1; If

the variable is 0, no warning about ill reaction events will be generated.

At the end of every simulation run the RS code writes a summary of the simulation run to the log file. It contains the total number of performed time steps, the mean time step length and the number of ill events, like for example:

```
#120 123.5950 ill events: 100
```

Despite the common data recording features of SIMION, RS provides the logging of the numbers of particles of the individual chemical species which allows to analyze the temporal development of the number concentrations of the chemical species. This feature is controlled by two adjustable variables: `RS_log_ts_period` controls the length of the period between the concentration logging events in time steps. If this variable is set to 0, the logging is disabled. The adjustable `RS_log_max_ts` controls how many times the concentrations are recorded.

The logged information has the following structure: Time-step number, passed time in simulation (in μs) and a comma separated list of the numbers of the particles according to the chemical species in the same order as they occur in the configuration file. For example the concentrations in the 94th time-step of a simulation with two species could be this:

```
94 , 358.7626 , 5937 , 4303 ,
```

Typically, the ions would be flown grouped with RS in SIMION because if they were flown ungrouped (i.e. one after another), only one ion would exist at a time in the simulation loop and so the logging of the number concentrations would be useless. However, if there are only independent reactions (which is actually enforced by RS) the ions can also be flown ungrouped if the logging of the number concentrations is not required.

3.2 RS standalone mode

The RS code can also be used in a standalone mode, without an ion trajectory simulation in

SIMION. In this mode, the model assumes an ideally stirred reactor, which means that in the world of the simulation there is no gradient (concentrations, temperature, pressure) at all ("one pot" mode). The standalone mode was initially introduced for testing and debugging purposes but it is also suitable for modeling problems where the assumption of an ideal stirred reactor holds true.

The standalone simulation mode is a commandline tool and requires a working Lua interpreter.

Notice: If SIMION is installed, you just click "Run Lua Program" on the SIMION main screen or use the command line methods such as the included `lua.exe` or `"simion.exe -nogui lua"`.

The wrapper script is named `RS_onePotSimulation.lua`. Its invocation is pretty straightforward:

```
lua RS_onePotSimulation.lua ↪
    nSimSteps maxDt ↪
    nParticles outputFileName
```

The commandline options are:

nSimSteps The number of simulation time steps. After this number of simulation time steps has passed, the simulation is terminated

maxDt The maximum possible length of a time step. The actual time steps are uniformly random distributed in the interval $[0, \text{maxDt}]$

nParticles The number of simulation particles, which are initially all created as particles of the first substance in the configuration file.

outputFileName The name of a result file in which the results of the simulation run are written.

The result of a standalone simulation is a simple text file, that contains the number concentrations for every time step of the simulation run. The information is given as one semicolon separated line per time step, for example:

```
4.529;728;2258;7014;
```

The first value is the time (in the same unit as the time unit given implicitly with the reaction rate constants), and the subsequent values are the number simulation particles of the simulated substances in the same order as their occurrence in the config file.

3.3 Config file documentation

RS config files are plain text files that consist of two parts: The definition of the substances in the simulation and the definition of the chemical reactions.

The `[substances]` keyword marks the substance definitions. Substances are given as a list, one substance per line. A substance line consists of the substance name, the type of the substance and a list of numerical parameters whose meaning depend on the substance type.

Notice: The substance name can not contain spaces and it can not start with a number, but it can contain numbers and underscores ("_").

Currently there are two types of substances:

discrete A substance, that is represented by discrete simulation particles. In the case of SIMION / RS, this would probably be an ionic substance. It has two parameters:

- The particle mass
- The particle charge

isotropic An isotropically distributed and static background substance. This substance type has only one parameter: the isotropic and static concentration, given in the same concentration units as used for the reaction rate constants.

For example, a definition of singly positive charged discrete substances A and B with masses of 30 amu and 40 amu respectively and an isotropic substance I1 with an concentration of $1 * 10^{10}$ would be as follows:

```
[SUBSTANCES]
A discrete 30 1
B discrete 40 1
I1 isotropic 1e10
```

The `[reactions]` keyword marks the reaction definitions. Every line in the reactions list defines an elementary reaction. The reactions are given in this format:

```
Educts=>Products ; rate-const
```

rate-const. is the rate constant of the reaction. In standalone mode the rate constant is given in simulation time units, which are arbitrary. The time unit of the rate constants decide which unit is the basic unit of time measurement in the simulation. In combination with SIMION the basic time unit is microsecond (μ s), but the rate constants are given with second (s) as basic time unit, and the rate constants are converted to microseconds as time unit by a conversion factor of 10^{-6} .

Educts and **Products** are multiplier / substance lists, which are given in the format:

```
multiplier substance-name + ↵
multiplier substance-name ↵
+ ...
```

The **multiplier** is an integer number; if it is omitted it is assumed to be one. The parameter **substance-name** is the name of a substance from the substance definitions. The multiplier / substance list can be of arbitrary length. It can (but should not for better readability) contain a distinct substance multiple times.

A valid definition of a simple reaction system (in this case a simple equilibrium) with the substances defined above would be:

```
[REACTIONS]
A + I1 => B ; 1E-10
B + I1 => A ; 2.5E-10
```


Notice: The RS configuration is not checked semantically and will be used if it is syntactically correct. It is up to the user to ensure that a RS configuration reflects a reasonable physical problem within the current restrictions of the RS implementation.

4 References / Literature

References

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